

AI Security Automation Platform - Detailed Report

1. Executive Summary

This report details the design and implementation of the AI Security Automation Platform, a comprehensive solution that integrates Data Ingestion, Threat Intelligence, Machine Learning Anomaly Detection, and SOAR orchestration.

2. Architecture & Modules

- Data Ingestion Layer: Simulates incoming network traffic logs and normalizes them into JSON formats.
- Threat Intelligence Enricher: Analyzes IPs and domains, enriching them with risk scores using simulated queries to AlienVault OTX/AbuseIPDB.
- Machine Learning Detection: A pre-trained Isolation Forest model that evaluates numerical features to assign anomaly scores.
- SOAR Engine: Orchestrates playbooks for High Confidence alerts (e.g. firewall blocking) and Medium Confidence alerts (e.g. ticketing).
- Visualization Dashboard: A real-time Streamlit dashboard that visualizes alerts and threat metrics continuously.

3. Demo Screenshots

Figure: p1.PNG

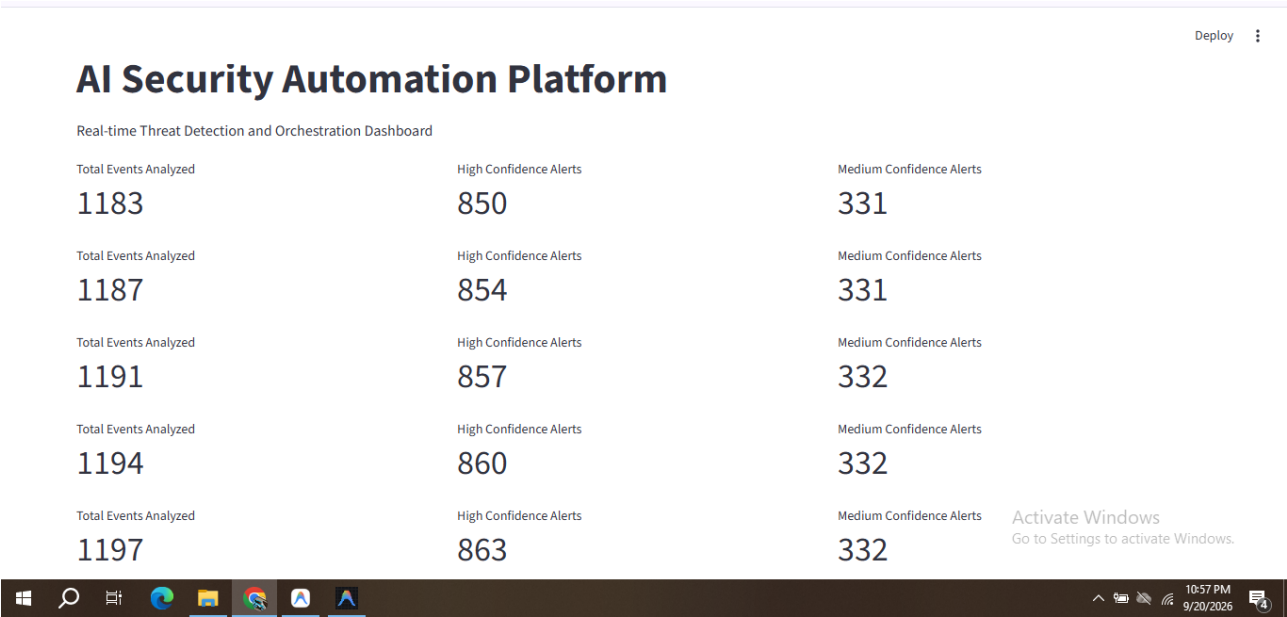


Figure: p2.PNG

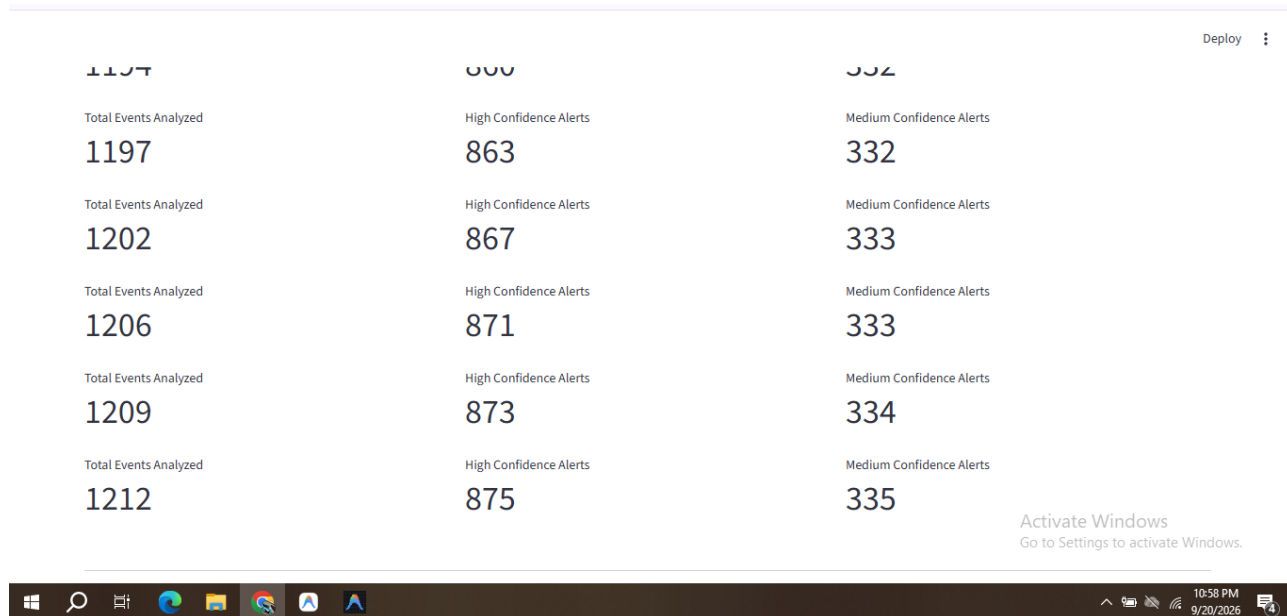


Figure: p3.PNG

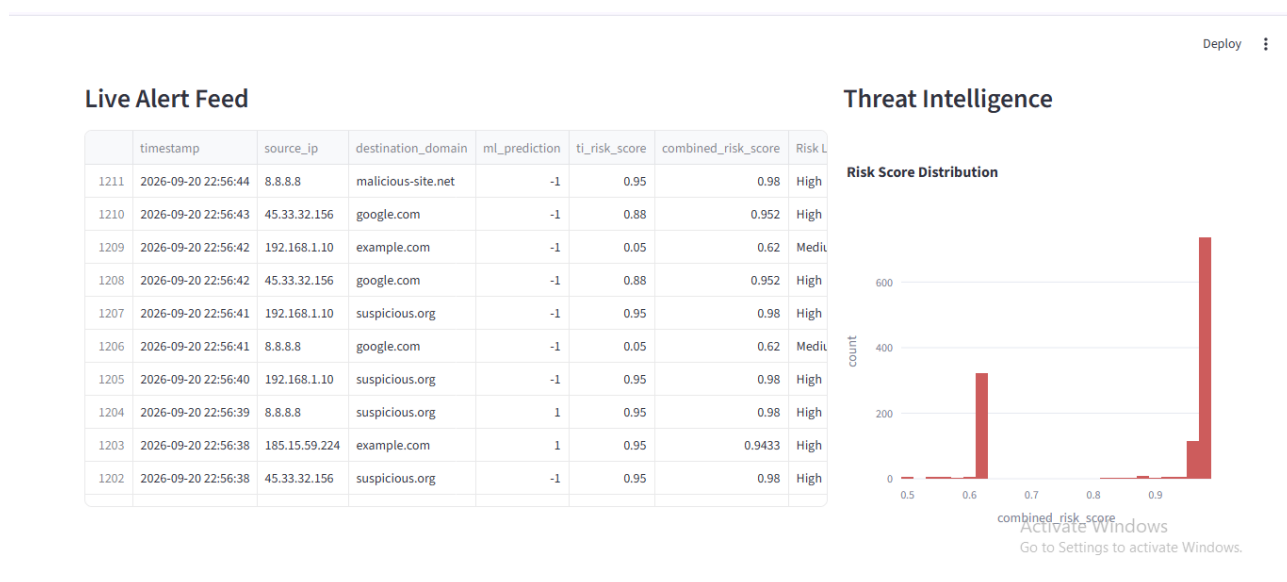
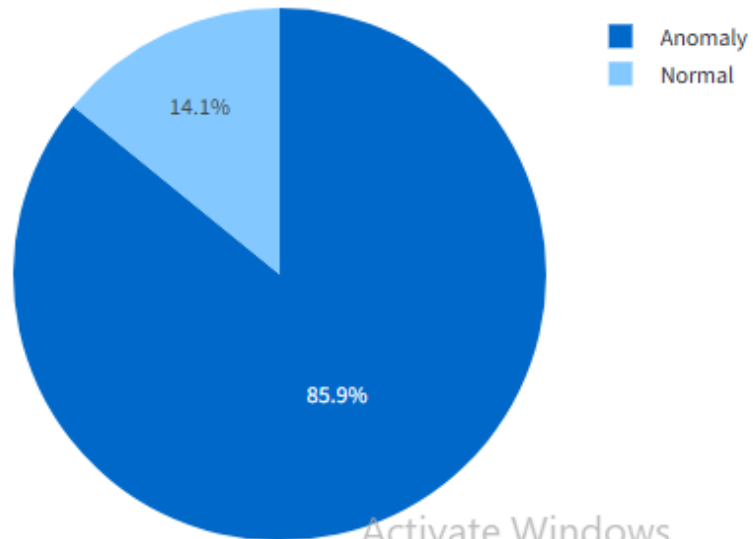


Figure: p4.PNG

Anomaly vs Normal

ML Detections



Activate Windows
Go to Settings to activate Windows.
